

Rushed to the altar: Social interactions of migrant workers and early marriage

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ABSTRACT: This paper provides novel evidence that social interactions of rural-to-urban migrants enforce traditional gender norms. Using variation of social pressure to conform to rural norms from migrants from the same hometown in the workplace, I find that the concentration of same-origin co-workers substantially increases the likelihood of early marriage for female rural-to-urban migrants in China, but not for male migrants. Consistent with the norm-based explanation for the association between social interactions and early marriage, the effect is more pronounced for migrants from regions with more traditional gender norms. The effect is not driven by matching or self-selection into social interactions.

Key words: social interactions, social norms, marriage, rural-to-urban migrants

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1. Introduction

Urbanisation has created unprecedented labour market opportunities for female rural-to-urban migrant workers. These opportunities have the potential to challenge traditional gender norms rooted in agricultural societies. Nevertheless, previous studies have found that migrants may continue to adhere to traditional gender norms despite exposure to new market forces and institutions at their destinations (e.g., [Fernández and Fogli, 2006](#); [Fernández and Fogli, 2009](#); [Blau, Kahn and Papps, 2011](#); [Charles, Guryan and Pan, 2022](#)).

This paper provides novel evidence on the persistence of traditional gender norms through social interactions and pressure. I use variation in social pressure to conform to rural norms from migrants from the same hometown in the workplace. Given the strong kinship and shared-origin ties, co-workers from the same hometown can exert greater pressure on migrant workers to conform to rural norms than those from other rural backgrounds. By examining rural-to-urban migrants, the effect of cultural norms can be isolated from that of rural occupations. Compared to friendship, co-workers are less subject to selection bias as individuals have less control over the origin of their co-workers and self-selection into co-workers, if any, is unlikely to be driven primarily by a preference for early marriage.¹ Because the social norm of early marriage is much more pronounced for females in agricultural societies than for males, I examine the effect separately by gender.

Using a discrete-time hazard model, I find that intensive social interactions (with more than 30% co-workers from the same hometown) pressure female migrant workers into early marriage, particularly increasing their probability of getting married before age 25 from 0.4 to 0.67.² In contrast, social interactions with same-origin co-workers have little effect on the distribution of marriage ages for male migrants. I show the robustness of the result using a simple linear probability model with an extensive set of controls including individual characteristics, job-related factors, and origin and destination fixed effects.³

Individuals may select into same-origin co-workers based on unobservable characteristics that reflect certain private preferences or tendencies for early marriage. A pertinent concern is that individuals with traditional values might have a preference for a workplace with same-origin co-workers and an inclination towards early marriage at the same time. I provide several pieces of evidence that this is unlikely to be the case. First, if the concentration of same-origin co-workers has no real impact on marriage behaviour but instead captures these private

¹As expected, the paper finds that same-origin co-workers do not pick up women's preference or tendency to marry early, whereas friendships with same-origin individuals are correlated with women's propensity to marry early.

²To put the number into perspective, extending the use of contraceptive pills to young unmarried women in the U.S. contributed to a reduction in the proportion of married college graduate women before age 26 from 0.7 for the cohort born in 1950 to 0.54 for the cohort born in 1957 ([Goldin and Katz, 2000](#)).

³The controls include individual characteristics such as educational attainment and birth cohort; job-related factors, including occupation, industry, salary, whether it is the individual's first job, whether the job was obtained through referral, and the age at which the individual started the job; as well as origin and destination province fixed effects.

preferences for early marriage, we would expect a correlation between early marriage and current same-origin co-workers present even for female migrants who were married prior to entering the workplace. Reassuringly, both the significance level and point estimate drop to zero when we examine females who have been married before migration.⁴ ⁵Second, while migrant workers' identification with their rural origin predicts early female marriage and serves as a plausible proxy for a preference or tendency for early marriage, it is not correlated with having more same-town co-workers. In the absence of random assignment of individuals to co-workers, I cannot definitively rule out selection on unobservables that, while unrelated to a preference for early marriage, may still influence its likelihood exclusively among female migrant workers. However, this possibility seems unlikely given the extensive controls for individual, job-related, and origin and destination factors. I therefore interpret my findings as strongly suggestive of a causal impact of social interactions on early female marriage, together with the various results presented throughout the paper that paint a consistent picture.

In line with the norm-based explanation, I find that the gender-differential effect of social interactions on early marriage is primarily concentrated among female migrants from regions with strong traditional gender norms. In other words, the effect of having more same-origin co-workers is more pronounced for individuals from regions with stronger traditional gender norms. In addition, examining a different outcome—whether the spouse works in the same workplace—reveals a similarly gendered effect of same-origin co-workers. Having more than 30% co-workers from the same hometown increases the likelihood that the spouse is in the same workplace, but this effect is much stronger among female migrant workers, and only among those originally from regions with stronger traditional gender norms. The finding is consistent with the idea that intensive social interactions exert pressure to conform to traditional norms, leading female migrants to be more likely to marry someone in the workplace, regardless of whether that person is from the same hometown. The gender-biased effects on earlier marriage and spouse also suggest that the effects of social interactions do not work through an increased likelihood of marriage match between men and women from the same hometown.⁶

The gendered effect of social interactions on early marriage echoes the gendered marriage norms rooted in agricultural societies. During China's prolonged agrarian era, early marriage was commonplace for females, reinforcing the formation and intergenerational transmission

⁴While it is possible that the share of same-origin co-workers may be salient to unmarried migrant women seeking to marry—if such networks facilitate matching—Section 6.1 shows that the concentration of same-origin co-workers did not result in matching.

⁵This also contrasts with the results concerning friendships formed in migrants' current location. Restricting the sample to individuals who married prior to migration (i.e., the same placebo test but using friendship as the explanatory variable), there is a positive association between early marriage and having friends from the same hometown in the individual's current location, suggesting that people may select into such friendships based on tendencies toward early marriage.

⁶Although men tend to marry later than women, the presence of same-origin co-workers does not significantly increase the likelihood of marriage at any age for men. Among females, the effect is concentrated in early marriages, whereas under a matching mechanism, one may expect effects more even distributed across different ages.

of this norm.⁷ According to the 2010 Chinese General Social Survey, people from rural areas, compared to urban dwellers, are more likely to approve of traditional gender roles, where men should prioritise their careers and women should prioritise family (Table A1). These norms, in turn, are reflected in behaviour. Individuals in rural areas tend to marry at an earlier age, particularly before age 25, compared to their urban counterparts. This distinction is more pronounced for females than for males (Figure A1). As individuals migrate from rural to urban China, the socio-economic environment that historically facilitated early marriage ceases to operate. Despite this change, I find that female migrants are more likely to marry early if they continue to identify with their rural origins. More importantly, this paper shows that the presence of same-origin migrants in the workplace pressures individuals to conform to traditional gender norms, regardless of their personal identification with their rural identity.⁸

This paper contributes to two strands of literature. First, it adds to the literature on the persistent effects of gender norms (e.g., Antecol, 2000; Fernández and Fogli, 2006; Fernández and Fogli, 2009; Blau, Kahn and Papps, 2011; Charles, Guryan and Pan, 2022).⁹ Traditional gender norms are strongly linked to agriculture (e.g., Alesina, Giuliano and Nunn, 2013; Hansen, Jensen and Skovsgaard, 2015), with idealised gender roles where men “work outside the home in the fields, while women specialised in activities within the home.” Previous research isolates the effect of culture by examining outcomes of migrants from different countries of origin—with varying social norms—who now live in the same country and therefore face similar market forces and institutions.¹⁰ This paper looks at individuals migrating internally from rural to urban China with varying exposure to same-origin co-workers to identify the persistence of rural norms through social pressure. By studying persistent gender norms in the context of internal migrants, the paper connects particularly to Charles, Guryan and Pan (2022). They separately identify the influence of sexism from one’s birthplace and one’s current location by analysing movers across U.S. states, which vary in their degrees of sexism. Consistent with their findings, I find that gender norms from individuals’ regions of origin continue to shape behaviour in the context of internal migration, and I highlight that their impact persists and varies with the social pressure to conform to these norms.

Second, this paper contributes to a small but growing body of literature on how women’s

⁷Historical records date back to the Western Zhou period (1046 BC–771 BC), during which the dynasty stipulated the maximum marriage age of 30 for males and 20 for females, requiring obligatory parental consent. During the reign of Emperor Hui (194 BC–188 BC), unmarried women between 15 and 30 years of age faced a tax equivalent to one year’s consumption of crops.

⁸The phenomenon in which individuals in a social group privately reject traditional norms yet continue to conform due to perceived peer pressure can be attributed to systematic misperceptions of social norms within peer groups, as shown by Miyajima and Yamaguchi (2017); Burszty, González and Yanagizawa-Drott (2020); Cortés, Koşar, Pan and Zafar (2024).

⁹See Giuliano (2021) for a review of the literature on gender and culture, and Olivetti, Pan and Petrongolo (2024) on social norms.

¹⁰In addition to gender norms, research examined the influence of culture on living arrangement (Giuliano, 2007), preferences for redistribution (Luttmer and Singhal, 2011) and levels of trust (Algan and Cahuc, 2010), among others.

expanding economic opportunities clash with traditional gender roles. Urbanisation offers unprecedented job opportunities for female migrants. Previous studies have shown that improved labour market opportunities for women are associated with delayed marriage and childbearing (e.g., [Jensen, 2012](#); [Heath and Mobarak, 2015](#)).¹¹ However, social interactions with co-workers from the same hometown pressure female migrants into early marriage—a choice that may no longer be optimal given their enhanced career prospects. This is closely related to the study by [Bertrand, Cortes, Olivetti and Pan \(2020\)](#), which finds heterogeneity in women’s responses to economic opportunities across different countries, measured by the marriage gap between skilled and unskilled women. They show that this heterogeneity is driven by society’s varying attitudes toward working women. The stickiness of gender norms challenges policy interventions aimed at tackling gender inequalities with economic incentives alone, as individual responses are mediated by conservative norms ([Olivetti, Pan and Petrongolo, 2024](#)).

The remainder of this paper is organised as follows. Section 2 introduces the institutional background of the emergence of a large number of migrant workers in China. Section 3 describes the data. Section 4 presents the estimation methods and results. Section 5 provides evidence for the underlying mechanism. Section 6 rules out alternative explanations. Section 7 concludes.

2. Institutional background

In this section, I provide an overview of the causes and consequences of rural-to-urban migrant workers in China.

The economic reforms of China since 1978 set in motion large-scale migration from rural to urban areas, driven by a surge in employment opportunities in urban manufacturing and service industries. Rural reforms also freed farmers from the land, further facilitating their migration into cities. In 2011, the number of migrant workers in China totalled 252 million, with 43.2% concentrated in the Pearl River and Yangtze River Deltas ([Figure A2](#)).¹²

The *hukou* system acts as an institutional catalyst that leads to the economic and social isolation of migrant workers in cities. This housing registration system mandates individuals to be registered under their place of origin and classified as either rural or urban residents, which, in turn, is used to link individuals to local public welfare programs. For rural-to-urban migrants, the conversion to local *hukou* remains extremely difficult ([Song, 2014](#)). Without *hukou* status in their cities of residence, migrant workers have limited access to local public goods, which reduces their bargaining power in the labour market ([Song, 2014](#)).¹³

¹¹ See also [Figure A1](#) for the estimated marriage hazard rates for the urban and rural populations using the sample from the 2010 Chinese General Social Survey.

¹² Source: the Chinese National Bureau of Statistics.

¹³ Studies find that rural migrants face severe labour market discrimination and the estimates range from 28% to 60% income differentials even after controlling for observable characteristics ([Liu, 2005](#); [Deng, 2007](#); [Gravemeyer, Gries and Xue, 2011](#); [Lee, 2012](#)).

Due to their disadvantaged positions in cities, most migrant workers take on jobs that are unappealing to local residents, characterised by long hours, poor working conditions, and low and unstable pay (Wang and Zuo, 1999). Migrant workers are also distinguished by their appearance and accent, a result of the pre-existing developmental disparities between rural and urban China, which further contributes to their social exclusion within cities.¹⁴ Migrant workers may stay close to their *tongxiang*—a Chinese term referring to people from the same hometown—to share networks, information, and resources. Occasionally, successful migrant workers rise to managerial positions, enabling them to hire others from the same hometown. This sometimes leads to a concentration of same-origin migrants within a particular workplace.

Intensive interactions with individuals from the same hometown can influence marriage behaviour through several channels. First, social interactions can increase the likelihood of finding a partner through increased opportunities for a marriage match. Additionally, social interactions can exert peer pressure to marry. China's rural communities are characterised by strong ties of kinship and shared origins, which often lead individuals to conform closely to the norms of their same-origin social group.¹⁵ This paper finds evidence for the latter channel—that social interactions influence migrant workers' marriage decisions through social pressure to conform to the gender norms of agricultural society.

3. Data

The main analysis of this paper is based on survey data from interviews of a cross-sectional 4157 rural-to-urban migrant workers in China in 2010.¹⁶ It took a representative sample of migrant workers working in the Yangtze and Pearl River Deltas in 2010.¹⁷ The quota sampling method was used to correct for representativeness in gender, occupation, and regional distribution of migrants, based on official statistical yearbooks. The data were used by Zhang and Xie (2016) to study the effect of the *tongxiang* network on migrant workers' wages. Consequently, in the 2010 wave of the survey, they incorporated questions regarding relations with *tongxiang*, or people from the same hometown, information that had not been covered in previous surveys in China.

¹⁴According to the 2010 survey on migrant workers, over 50% of the migrant workers reported discrimination by local residents and approximately 60% lacked of a sense of belongingness in the cities.

¹⁵While co-workers or friends from the same town or county may not be biologically related to an individual, they can report that individual's behaviour to the closely connected rural community back home. This places the individual's behaviour under the scrutiny of their same-origin co-workers.

¹⁶The survey was part of a research project sponsored by the Ministry of Education in China with the aim of studying the status-quo of migrant workers and protecting their rights.

¹⁷According to the Chinese National Bureau of Statistics, the two regions have the highest concentration of migrant workers in China and together assimilated over 40% of total migrant workers in 2011. Figure A2 depicts the migration outflows to the Pearl and Yangtze River Deltas based on the same survey data.

3.1 Measures of social interactions

The survey data contain information regarding the percentage of co-workers from the same province, county, and town, which are respectively first-, third- and fourth-level administrative units in China. In addition, we know whether the migrant worker's best friends in their current location are from the same hometown, as identified by the respondents. In the survey, individuals were asked the following question:

Q1 On the production line (team) that you work, what is the proportion of people from the same town, county, or province?

(1) None, (2) Very few (<10%), (3) A few (10%–20%), (3) Some (20%–30%), (4) Many (30%–50%), (5) A lot (> 50%), (6) Do not know.

Table 1 reports the percentage of same-origin co-workers in the same production line/team, defined by co-workers from the same town, county, or province, respectively. A sizeable proportion of people have over 30% of co-workers from the same region. Table A2 reports the proportions of same-town co-workers by gender and by marital status prior to migration.

Table 1: Proportion of *tongxiang* co-workers

	None	Very few <10%	A few 10%–20%	Some 20%–30%	Many 30%–50%	A lot >50%
Town	0.483	0.265	0.091	0.072	0.040	0.050
County	0.412	0.265	0.117	0.091	0.053	0.061
Province	0.140	0.201	0.137	0.137	0.153	0.233

Notes: Based on the 2010 survey on migrant workers. The table presents the levels of the concentration of co-workers (in columns) from the same town, county, and province (in rows).

The concentration of same-origin co-workers is my preferred measure of social interactions because it is a less endogenous choice than friendships. Moreover, the concentration of co-workers is a good proxy for social interactions as migrant workers spend a substantial amount of time with their co-workers. According to the same survey, the average working hours are 9.3 hours per day on an average 6-day working schedule. In addition, 32% of total migrant workers even work 7 days a week. The intensity of the working schedule results in the intensity and closeness of social interactions within the workplace. Migrant workers have numerous opportunities to socialise (Fang, 2012). For example, 36% of total migrant workers live in the dormitory provided by their employers and the percentage increases to 49% for individuals who are single; 62% of workers dine in the canteen of the workplace.

Another potential measure of social interactions is migrant workers' self-perceived friendship with *tongxiang*. In the survey, respondents were asked to identify three of their best friends in their current location and whether each friend is from the same hometown. The data indicate that a considerable proportion of people have *tongxiang* as their best friends. The proportion of

individuals whose first best friend is *tongxiang* is 0.37. If we consider three best friends, for 13% of the individuals, all their best friends are *tongxiang*, and for 44% of the individuals, at least one of their best friends is *tongxiang*.

Friendships are more susceptible to selection bias since people choose their friends. Specifically, individual with more traditional values may be more likely to form friendships with *tongxiang*. On the other hand, individuals have less control over the composition of their co-workers, and even if they do, such selection is unlikely to be driven by a preference for early marriage, conditional on observable individual characteristics. In fact, the paper shows that *tongxiang* co-workers are not correlated with a preference for early marriage, while *tongxiang* friendships are associated with such a preference. Therefore, the main analysis focuses on the concentration of same-origin co-workers to study the effect of social interactions on marriage age.

3.2 Individual characteristics

The sample consists of 1895 females and 2252 males. After excluding observations that are either widower or divorced, a sample of 4093 individuals remains.

The average migrant worker in the sample has 9 years of education and migrated at an age of 19. The migration is typically temporary. Less than 25% migrant workers express a desire to convert to local *hukou*, not to mention the practical barriers to acquire one. The majority of individuals work in manufacturing and service industries with an almost equal number of male and female workers. There are significantly more males in construction than females although the total employment is only 308, which also makes up for the difference between the total number of females and males in the sample.¹⁸ Transportation assimilates the fourth largest employment with a balanced gender representation. The other sectors are relatively small and assimilate approximately 2% of workers (see Figure A3).

3.3 Complementary data

I use the 2010 Chinese General Social Survey which enquires about individual attitudes towards gender roles in society to measure regional variation in the strength of traditional gender norms. If social interactions with same-origin co-workers pressure females into early marriage through cultural norms, this effect is likely to be stronger among individuals from regions with more traditional gender norms.

4. Estimation methods and results

In this section, I present the estimation methods and results. First, I use a discrete-time duration model to examine the effect of social interactions on the distribution of marriage age, quantifying

¹⁸Estimation results in Section 4 are robust to exclusion of the construction sector.

the effect by translating the hazard rates into cumulative distribution functions. The analysis shows that female migrants are more likely to marry at an early age, particularly before 25, if their co-workers are from the same town. In contrast, same-origin co-workers have little impact on the distribution of marriage age for male migrants. Then, I show that the gender-differential effect of social interactions is robust when using a linear probability model specification, which controls for an extensive set of individual characteristics.

4.1 Estimation of hazard rate

I examine the effect of social interactions on the probability of getting married at different ages. While one could alternatively estimate this effect by comparing average marriage ages, such a comparison would overlook individuals who remain unmarried in the sample. The absence of marriage, particularly among those without same-origin co-workers, also provides valuable information into the impact of social interactions on marriage. Additionally, analysing the probability of marriage at different ages, rather than focusing solely on average marriage age, can provide more detailed insights into the underlying mechanisms at play.¹⁹

Hence, I employ a duration framework to model the probability of getting married at different ages. This approach incorporates information on individuals who have not yet married within the sample. It also facilitates a detailed examination of the effect of social interactions on the entire distribution of marriage age, beyond just the mean. Following the specification of the logistic discrete-time duration model by [Bover, Arellano and Bentolila \(2002\)](#), I define the hazard rate of marriage $h(t)$ as a function of the natural logarithm of age t .

$$h(t) = Pr(T_i = t | T_i \geq t) = G(\gamma_0 + \gamma_1(\ln t) + \gamma_2(\ln t)^2) , \quad (1)$$

where

$$G(x) = \frac{e^x}{1 + e^x} . \quad (2)$$

The duration here is years to marriage. The hazard rate $h(t)$ is the conditional probability of getting married at age t , given that one has not been married before age t . The specification here allows the hazard rate to vary with age t .

I examine marriage hazard rates at different ages and also examine how they vary with the concentration of co-workers from the same hometown, separately for male and female migrant workers. Panel (a) of Figure 1 shows the estimated hazard rates for female migrant workers with different levels of concentration of co-workers from the same town (the fourth-level administrative unit in China). As the proportion of same-town co-workers increases, there is a shift and intensification of exit rate from singlehood at early ages. The probability of marriage is

¹⁹For instance, if social interactions increase the probability of marriage across all ages, this pattern is more consistent with a matching mechanism. In contrast, if the effect is concentrated at younger ages, it more likely reflects social norms.

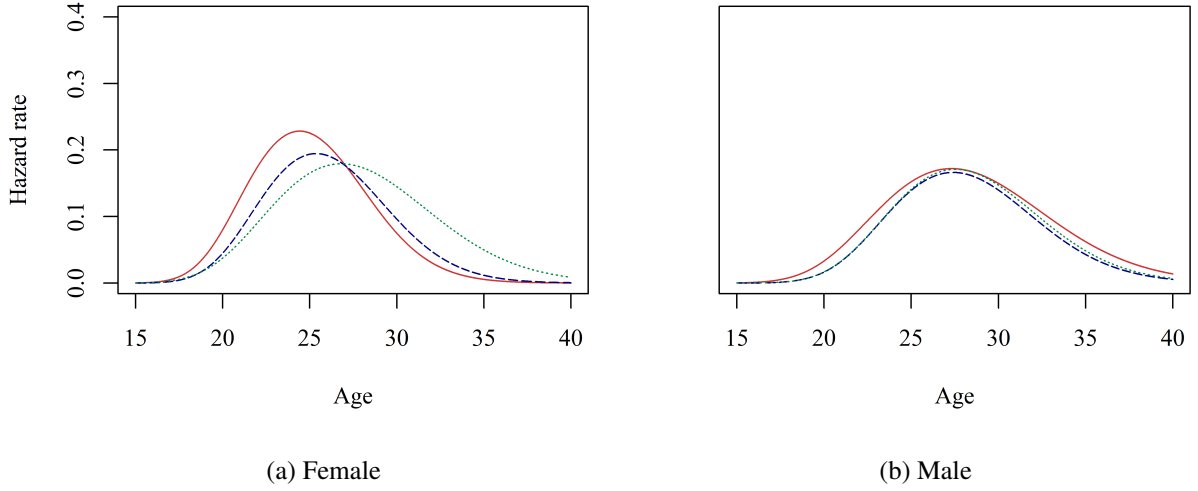


Figure 1: Estimated hazard rate: pc_town

- none of the co-workers come from the same town
- - - 0–30% of the co-workers come from the same town
- >30% of the co-workers come from the same town

particularly heightened just before the age of 25. This indicates that female migrant workers are more likely to marry early when a higher proportion of co-workers are from the same hometown. In contrast, the concentration of same-town co-workers does not appear to affect the distribution of marriage age for male migrant workers, as shown in the right panel of Figure 1.

Similar gender-differentiated patterns emerge when examining co-workers from the same county, and the results are presented in the Appendix (Figure A4). The effect of co-workers from the same province is smaller (Figure A5). Given that the average area of a province in China is comparable to that of Germany, and its average population is similar to that of Spain, there is substantial heterogeneity within a province. Consequently, individuals may not always identify people from the same province as *tongxiang*.²⁰

4.2 Estimation of cumulative distribution function

The estimated hazard rates offer an intuitive overview of the effect of social interactions on marriage age. To quantify this effect, I translate the hazard rate into a cumulative distribution function, calculated as follows:

$$F(t) = 1 - \prod_{t=1}^T (1 - h(t)) , \quad (3)$$

where $F(t)$ represents the probability that an individual gets married at or before age t . This probability can be compared between individuals with more or less same-origin co-workers.

²⁰The average area of a province in mainland China is 352,033 km², excluding provincial level municipalities (i.e., Beijing, Shanghai, Tianjin, and Chongqing). The average population of a province was 46 million in 2010.

Table 2: Marriage age and same-town co-workers—main analysis

Age	Probability of being married at or before a certain age					
	Female			Male		
	Baseline 0 same origin (1)	Difference >30% (2)	Difference >50% (3)	Baseline 0 same origin (4)	Difference >30% (5)	Difference >50% (6)
20	0.070	0.069* (0.039)	0.101 (0.079)	0.027	0.035* (0.019)	0.046 (0.025)
21	0.128	0.122** (0.048)	0.167** (0.081)	0.060	0.054** (0.026)	0.067* (0.032)
22	0.207	0.174*** (0.057)	0.227*** (0.084)	0.114	0.072** (0.031)	0.084** (0.038)
23	0.302	0.208*** (0.065)	0.261*** (0.089)	0.190	0.083** (0.036)	0.093** (0.046)
24	0.404	0.217*** (0.071)	0.265*** (0.093)	0.283	0.086** (0.041)	0.092* (0.054)
25	0.503	0.204*** (0.077)	0.245** (0.095)	0.385	0.081* (0.046)	0.082 (0.063)
26	0.591	0.177** (0.081)	0.212** (0.097)	0.483	0.071 (0.049)	0.069 (0.068)
27	0.664	0.145* (0.083)	0.174* (0.098)	0.571	0.060 (0.050)	0.056 (0.069)
28	0.722	0.114 (0.084)	0.139 (0.099)	0.644	0.050 (0.050)	0.046 (0.068)
29	0.767	0.087 (0.084)	0.109 (0.099)	0.701	0.042 (0.049)	0.038 (0.065)
30	0.801	0.064 (0.084)	0.084 (0.100)	0.745	0.037 (0.047)	0.034 (0.062)

Notes: The estimation is based on the sample of individuals married after migration. The baseline group is migrants without any co-workers from the same town. The difference is the increase in probability relative to the baseline group for migrants with over 30% or 50% of the co-workers from the same town. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table 2 compares individuals with over 30% or 50% of their co-workers from the same town to those without any co-workers from the same town. The sample is restricted to individuals who get married after migration. For presentation purposes, the table only reports the probability of getting married between the ages of 20 and 30.²¹ In columns (1) and (4), I show the probability of getting married at or before a certain age for the baseline group of female and male migrants, respectively. The baseline group comprises migrant workers who have no co-workers from the same hometown. The differences in columns (2) and (3) for females, and columns (5) and (6) for males, yield the effect of social interactions, which is the increase relative to the baseline group in the probability of getting married for migrant workers with more than 30% and 50% of

²¹The probability of marriage can be calculated for all ages starting from 16, which is set to be the minimum marriage age.

their co-workers from the same hometown. Bootstrap standard errors with 500 replicates are reported in parentheses.²²

A female migrant without any same-town co-workers has a 21% probability of being married at or before age 22 (column 1). If the majority of her co-workers are from the same hometown, this likelihood increases by 23 percentage points (column 3). In contrast, the effects on male workers are significantly smaller (columns 5 and 6). The largest effect for females is observed at age 24, specifically in the probability of being married before 25. For female migrants whose majority of co-workers are from the same hometown, this probability increases by 26.5 percentage points.

As a placebo test, we can make the identical calculation for individuals who have already been married before migration. If social interactions have no real impact on marriage behaviour, but instead capture certain tendencies for early marriage, we would expect a correlation between early marriage and social interactions with same-origin co-workers even among females who were already married before entering a workplace with same-town co-workers.²³ In Table A3, I estimate the effect of current same-town co-workers, but look at individuals who were already married before migrating into a city. Now, the effect of social interactions disappears for females (columns 2 and 3). Not only does the significance level fall to zero but so do the point estimates. This suggests that social interactions have a real impact on marriage behaviour rather than merely reflecting spurious correlations.²⁴

In Table A6, I present additional results on the effect of social interactions with 0–10%, 10–30% co-workers from the same town, juxtaposed with the effect of having over 30%. The results show that the effect gradually increases with a higher proportion of same-town co-workers. The effect of having 10–30% same-town co-workers on early marriage is positive and statistically significant, though smaller in magnitude compared to the effect of having over 30%. These findings indicate that the observed effect of intensive social interactions—such as with more than 30% or 50% same-town co-workers—is unlikely to be driven by random chances from a particular subsample.

The estimation results using same-county co-workers (Table A7 and A8) show similar patterns: (1) female migrant workers with same-origin co-workers are more likely to marry early; (2) the effect is strongest for marriages occurring before age 25; (3) the effect is much

²²The estimation of duration model generates $t-15$ (the minimum marriage age is set to be 16) observations for individuals who get married at age t and generates the number of observations that equals one's age minus 15 if the individuals is not married. The bootstrap error is calculated by clustering at the individual level. Clustered bootstrapping is equivalent to taking a random sample of individuals from the original sample (with replacement) and generates the above-mentioned expanded sample each time.

²³While it is possible that the share of same-origin co-workers may be more important for unmarried migrant women seeking to marry—if such networks facilitate matching—Section 6.1 shows that the concentration of same-origin co-workers did not result in matching.

²⁴Note that for male migrant workers who were married before migration, the association between social interactions and early marriage is slightly positive but very similar to the results for migrants who marry after migration. This implies positive sorting for male migrants. A closer look reveals that this relationship is primarily driven by the negative correlation of education with both same-origin co-workers and early marriage.

smaller for male migrants; and (4) the association between same-origin co-workers and early marriage is absent for female migrants who were married prior to migration.

By comparison, Tables A4 and A5 present estimates using the friendship measure. The gender-differentiated effects of social interactions are similar to those observed with co-workers. For a female migrant worker who marries after migration, she is more likely to marry early if her best friend in her current location is from the same hometown. The effect is smaller for male migrants.²⁵ Interestingly, even among individuals who were married before migration, there is a positive association between early marriage and having same-origin best friends for female migrant workers. This suggests that, unlike with co-workers, friendships with *tongxiang* are subject to selection in the same direction as early marriage for female migrants. In other words, female migrants with a tendency toward early marriage are more likely to form friendships with other migrants from the same hometown in their current location.

4.3 Estimation of linear probability model

The analysis using the discrete-time hazard model reveals that a higher concentration of co-workers from the same hometown increases the probability of early marriage for female migrant workers, especially before the age of 25. This finding aligns with the social pressure faced by many rural-to-urban female migrants to conform to traditional gender roles in agricultural societies, where age 25 often serves as a pivotal reference point for marriage. Figure A1 illustrates the same pattern using a different data set—the 2010 Chinese General Social Survey—by showing the hazard rates of marriage at different ages for both rural and urban populations. Among the rural population, hazard rates intensify notably at ages 24 and 25 for females, consistent with age 25 serving as a benchmark for marriage among women in rural communities.

In this subsection, I show the robustness of the gender-differential effect of same-origin co-workers using an alternative specification. Specifically, I employ the linear probability model to predict marriage before age 25.²⁶ Table 3 presents the estimation results. Column (1) shows the effect of having more than 30% of co-workers from the same town on female migrants without any controls. The results indicate a similarly 25.2 percentage point increase (21.7 in the discrete-time duration model) in the probability of being married before age 25.

²⁵Note that the estimated effect of best friends from the same hometown is smaller than the effect of having 50% of co-workers from the same hometown and is comparable to the effect of having 10% of co-workers from the same hometown.

²⁶The linear probability model treats individuals who are unmarried and under the age of 25 as missing, since we do not know whether they will marry before age 25. In contrast, the hazard model can still use these observations to estimate the hazard rates. The similarity in the effect of social interactions on the probability of marrying before age 25 across the two methods suggests that individuals for whom we do not observe the entire marriage history are not systematically different from those for whom we do.

Table 3: Married before age 25 and same-town co-workers

	Dep. var.: 1 (Married before age 25)					
	Female			Male		
	(1)	(2)	(3)	(4)	(5)	(6)
Same-town co-worker	0.252*** (0.069)	0.231*** (0.078)	0.162* (0.089)	0.050 (0.053)	0.057 (0.057)	−0.071 (0.060)
Education			−0.170*** (0.041)			−0.103*** (0.027)
Age at start of work			−0.014 (0.009)			−0.020*** (0.005)
First job			−0.123 (0.089)			−0.134** (0.058)
Referral			−0.010 (0.059)			0.034 (0.058)
Log(salary)			−0.121 (0.087)			0.011 (0.053)
Occupation FE			✓			✓
Industry FE			✓			✓
Birth cohort FE			✓			✓
Origin and destination FE			✓			✓
Observations	334	283	283	617	536	536
R-squared	0.028	0.024	0.391	0.001	0.002	0.193

Notes: The estimation uses a linear probability model to predict whether an individual marries before age 25. The sample includes individuals who married after migrating and either have no co-workers from the same town or have more than 30% of their co-workers from the same town. *Same-town co-worker* is an indicator for having over 30% of co-workers from the same town. *Education* is a categorical variable that increases with the level of education (0: less than or equal to primary school; 1: junior high school; 2: senior high school; 3: more than high school). *Age at start of work* is the age at which the individual began working in their current job. *First job* is an indicator for whether the current job is the individual's first job. *Referral* indicates whether the job was obtained through a referral. *Log(salary)* calculates the natural logarithm of the individual's monthly salary. Robust standard errors are given in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Column (3) includes an extensive set of controls, encompassing individual characteristics such as educational attainment and birth cohort, as well as job-related factors, including occupation, industry, salary, whether it is the individual's first job, whether the job was obtained through referral, and the age at which the individual started the job. Additionally, I include origin and destination province fixed effects. The positive effect remains for female migrants but shows a smaller 16.2 percentage point increase in the probability of being married before age 25.²⁷ The sample size in column (3) is smaller than in column (1) due to missing information on industry for some individuals. In column (2), I estimate the effect without controls using the same sample as in column (3), and the results show a very similar effect to that in column (1).

²⁷While the overall effect appears somewhat weaker after controlling for covariates, Section 5 shows a particularly strong effect among individuals from regions with traditional gender norms.

Columns (4)-(6) present the estimated effects for male migrant workers. Consistent with Table 2, the effect of having same-town co-workers on the probability of being married before age 25 is much smaller for male migrants. While men generally marry later than women, Table 2 also shows that this effect is absent for later ages among male migrants.

The results in this section, using both the hazard model and the linear probability model, indicate that a higher concentration of co-workers from the same town increases the likelihood of early marriage among female rural-to-urban migrants. This relationship does not appear to be driven by spurious correlations, unlike the patterns observed in friendships. The next sections explore the underlying mechanism.

5. Effect of social interactions through traditional gender norms

Anecdotal evidence suggests that female migrant workers in the workplace experience considerable social pressure to marry from their co-workers. Fang (2012) vividly pictures social interactions within an electronic factory in Shenzhen, Pearl River Delta:

On the factory floor, women have plenty chances to compare themselves to one another...Married women gossip aggressively about the unmarried, especially the “old” ones, and the gossiping does not take into account any career performance. No matter how well an unmarried woman performs in her job or how high the job position she attains, she will still be singled out for her embarrassing unmarried status. In other words, she has simply “failed” to marry.

Co-workers from the same hometown can exert much stronger social pressure than those from other equally rural origins. Relationships with same-origin co-workers tend to be closer, and these co-workers can also report individual behaviour back home, making the traditional norms more salient or enforceable.

I provide two sets of evidence to support the idea that the effect of social interactions on early marriage operates through social pressure to conform to the traditional gender roles in agricultural society. First, I show a gender-differentiated response to rural identity in the age at marriage. Female migrants who identify more strongly with their rural origin are significantly more likely to marry early, whereas the association between rural identity and early marriage is much weaker for male migrants. This suggests that the rural norm of early marriage, proxied by self-identification with a rural origin, is more binding for females than for males. Importantly, self-identification with a rural origin is not correlated with the concentration of same-origin co-workers, which also suggests that there is no selection into *tongxiang* workplaces based on a preference for early marriage.

Second, I show that the effect of same-origin co-workers on early marriage is mainly concentrated among individuals from regions with conservative views on women's roles in society. It is important to note that this second test should not be confused with selection into

same-origin co-workers among individuals from more traditional regions. Rather, it examines whether, with the same concentration of same-origin co-workers, its effect is stronger when traditional norms are more pronounced, as captured by an interaction between the presence of same-origin co-workers and a measure of norm strength. The correlation between hailing from a traditional region and having same-origin co-workers, or sorting, will be accounted for by including both variables separately in the regression.

5.1 Farmer v.s. Worker Identity

One's identity prescribes how people belonging to that group should behave ([Akerlof and Kranton, 2000](#)). Thus, migrant workers' identification with their rural origins subjects them to the social norms of agricultural society, which may continue to shape their behaviour even after migrating to cities. In this section, I explore this possibility as supporting evidence that the gendered effect of social interactions arises from gendered marriage norms in agricultural society.

Although less than 1% of individuals in the sample work in the agriculture sector in cities, many migrants still identify as farmers due to their rural origins and rural *hukou*. In the same 2010 survey of migrant workers, individuals were asked the following question:

Q2 Which of the following do you believe to be your identity?

(1) Farmer, (2) Worker, (3) Others, (4) Do not know.

Table 4 calculates the probability of getting married at or before a certain age by self-identity, separately for males and females, and only for individuals who get married after migration. The baseline groups are migrant workers who identify themselves as workers for females in column (1) and for males in column (3). I show the increase relative to the baseline in the probability of getting married for individuals who identify themselves as a farmer for females in column (2) and for males in column (4). Female migrants who identify themselves as farmers are more likely to get married early (column 2). For males, the association between rural origin and marriage age is much weaker (column 4). Similar to the effect of same-origin co-workers, the impact of self-identity is strongest at age 24, or in the probability of being married before 25, for female migrants.

One concern is that individuals who strongly identify with their rural origin may be more likely to sort into workplace with same-origin co-workers. As a result, the correlation between the prevalence of same-town co-workers and early marriage may instead reflect the effect of self-identity on marriage behaviour. In Table A10, I regress the indicator for self-identification as a farmer on the concentration levels of same-town co-workers with and without controls and find little correlation between the two.

Interestingly, among individuals who marry before migration, females identifying with their rural origins (as responded in the survey after migration and marriage) also tend to marry early, whereas their male counterparts do not (Table A9). This supports the idea that self-identity

Table 4: Marriage age and identity

Age	Probability of being married at or before a certain age			
	Female		Male	
	Baseline Worker (1)	Difference Farmer (2)	Baseline Worker (3)	Difference Farmer (4)
20	0.076	0.035 (0.029)	0.035	0.003 (0.011)
21	0.135	0.060 (0.037)	0.070	0.009 (0.018)
22	0.214	0.086* (0.045)	0.124	0.020 (0.026)
23	0.307	0.105** (0.053)	0.197	0.034 (0.034)
24	0.406	0.113* (0.058)	0.283	0.048 (0.041)
25	0.502	0.108* (0.062)	0.376	0.059 (0.045)
26	0.588	0.095 (0.063)	0.466	0.065 (0.047)
27	0.660	0.077 (0.063)	0.547	0.067 (0.047)
28	0.718	0.058 (0.064)	0.616	0.064 (0.045)
29	0.763	0.041 (0.065)	0.672	0.059 (0.044)
30	0.798	0.026 (0.067)	0.716	0.053 (0.044)

Notes: The estimation is based on the sample of individuals married after migration. The baseline group is individuals who identify themselves as workers. The difference is the increase in probability relative to the baseline group for migrants who identify themselves as farmers. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

remains stable over time. Female migrants who identify with their rural origins act according to gender norms prescribed by their rural identity, leading to early marriage. For males, however, the rural norm of early marriage is not as strong, so self-identification with a rural origin does not result in early marriage. Notably, the effect of farmer identity is weaker than the effect of same-origin migrants in the workplace. This may be due to either (1) farmer identity being a proxy for the social norm of early marriage, or (2) the intensifying and multiplying effects of social pressure on marriage behaviour.

5.2 Traditional v.s. Non-traditional Provinces

If rural norms pressure females into early marriage in the presence of same-origin co-workers, we would expect a stronger effect of social pressure when the norm is stronger. The strength of

the norm can be proxied by views regarding women's role in society. I use the response in the 2010 Chinese General Social Survey to the following question:

Q3 Do you agree that men should prioritise their careers and women should prioritise their families?

(1) Completely disagree, (2) Disagree, (3) Indifferent, (4) Agree, (5) Strongly agree, (6) Do not know.

I divide provinces into two equal groups and code them as traditional and non-traditional provinces depending on the proportion of rural respondents who agree or strongly agree with the above proposition (high proportion defined as traditional province, see Figure A6 for the classification). I classify provinces as traditional or non-traditional along this dimension because the question reflects the value placed on family over career for females, which closely aligns with preferences for early female marriage. I then compare the gendered effects of social interactions on early marriage between individuals from traditional and non-traditional provinces.

I investigate whether the effect of social interactions on early marriage is heterogeneous in Table 5. Similar to Table 3, I predict whether an individual is married before age 25 using a linear probability model. In addition to the indicator for having more than 30% same-town co-workers, I include an interaction term between same-town co-workers and an indicator for being from a province with traditional gender norms. Column (1) shows that having more than 30% same-town co-workers and being from a traditional province are both associated with a higher probability of being married before age 25 among female migrants, although the effects are not statistically significant. The coefficient on the interaction term is large and statistically significant, showing that the influence of same-town co-workers on early marriage likelihood is substantially strong for individuals from provinces with traditional gender norms. In column (2), I include the extensive set of controls. Now the direct effect of being from a traditional province is absorbed by the fixed effects for the origin provinces. The coefficient on same-town co-workers captures the effect for individuals from non-traditional provinces and the estimate is close to null. The interaction term captures the additional effect of same-town co-workers for individuals from provinces with traditional gender norms. The coefficient estimate indicates that if a female migrant hails from a province with traditional gender norms, the presence of same-town co-workers increases the probability of being married before 25 by approximately 40 percentage points. Turning to male migrant workers in column (3), there is a positive association between same-town co-workers and early marriage for individuals from non-traditional provinces. However, this association disappears once controls are included in column (4). Moreover, there is no additional effect of same-origin co-workers on early marriage among males from provinces with more conservative views on women's roles in society.

Table 5: Married before age 25 and same-town co-workers: Heterogeneity

	Dep. var.: $\mathbb{1}$ (Married before age 25)			
	Female		Male	
	(1)	(2)	(3)	(4)
Same-town co-worker	0.102 (0.125)	−0.025 (0.146)	0.086 (0.095)	−0.029 (0.095)
Same-town co-worker $\times$ Traditional	0.273** (0.137)	0.420** (0.175)	−0.062 (0.115)	−0.063 (0.114)
Traditional	0.072 (0.060)		0.073* (0.043)	
Controls		✓		✓
Occupation FE		✓		✓
Industry FE		✓		✓
Birth cohort FE		✓		✓
Origin and destination FE		✓		✓
Observations	334	283	617	536
R-squared	0.047	0.406	0.006	0.194

Notes: The estimation uses a linear probability model to predict whether an individual marries before age 25. The sample includes individuals who married after migrating and either have no co-workers from the same town or have more than 30% of their co-workers from the same town. *Same-town co-worker* is an indicator for having over 30% co-workers from the same town. *Traditional* is an indicator of whether an individual hails from a province with traditional gender norms, as determined by responses to Q3. The control variables are defined in the same manner as in Table 3 and they are educational attainment, log salary, an indicator for whether it is the individual's first job, an indicator for whether the individual was referred to the job, the age at which the individual started the job. Robust standard errors are given in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

The findings suggest that the effect of same-origin co-workers on early female marriage is driven by traditional gender norms. Female migrant workers from provinces with traditional gender norms are more likely to marry before the age of 25 when surrounded by a substantial share of same-town co-workers, compared to female migrants from similar provinces without same-town co-workers. In contrast, the effect of same-origin co-workers is considerably smaller for females from non-traditional provinces, as well as for males more generally, among whom early marriage norms are less strongly held.

6. Alternative explanations

In this section, I rule out alternative mechanisms that can generate the gendered effect of social interactions.

6.1 Matching

Migrant workers originating from the same hometown often share customs, dialects, and similar socio-economic backgrounds. The concentration of same-origin co-workers can create

opportunities for individuals to connect with others who share these similarities, thereby increasing the likelihood of meeting a potential partner and facilitating a quicker transition into marriage.

However, matching alone cannot account for the impact of social interactions on marriage, as the effect is much more pronounced for female migrants than for their male counterparts. While it is true that men generally marry later than women, we do not observe a comparable effect of same-origin co-workers on male migrants at later ages either. Moreover, the effect primarily concerns early marriage, particularly before the age of 25, rather than an increased likelihood of marriage across all ages. This aligns more closely with social pressure to conform to traditional gender norms. If the mechanism were better matching, the effect would likely be more evenly distributed across all age groups.

In addition, I explore a testable implication of the matching story. If workplace matching takes place, the likelihood of a migrant worker's spouse being in the same workplace may increase with a higher fraction of same-origin co-workers. To investigate this, I regress the indicator of the spouse being in the same workplace on the indicator for having more than 30% same-town co-workers, separately for male and female migrant workers who married after migration. The results are presented in Table 6. Column 1 indicates that when female migrants have more than 30% of co-workers from the same town, the probability that their spouse works in the same workplace increases by 19.5 percentage points. In columns (2), I further interact the indicator for same-town co-workers with an indicator for originating from a province with traditional gender norms. The direct effect of coming from a traditional province is absorbed by the origin province fixed effect. After including the interaction term, the coefficient of same-town co-workers drops to zero, capturing the effect of having more than 30% of co-workers from the same town for female migrants from non-traditional provinces. The interaction term captures the additional effect of same-origin co-workers on female migrants from traditional provinces, showing an approximately 47 percentage point increase in the likelihood of having spouse in the same workplace.

However, comparable effects are not observed among male migrant workers (column 3 and 4). If matching between same-origin co-workers took place, similar effects would be expected for both male and female migrant workers.²⁸ Note that an alternative scenario, where female migrants seek and work in their husband's workplace—potentially leading to more same-origin co-workers—would also be inconsistent with the results.²⁹ The gender-asymmetrical effect instead indicates that female migrants with co-workers from the same town are more likely to marry someone from their workplace, regardless of whether that person is from the same

Table 6: Spouse workplace and same-town co-workers

	Dep. var.: 1 (Spouse in the same workplace)			
	Female		Male	
	(1)	(2)	(3)	(4)
Same-town co-workers	0.196** (0.081)	−0.011 (0.102)	0.110* (0.060)	0.054 (0.085)
Same-town co-workers × Traditional		0.470*** (0.156)		0.081 (0.106)
Observations	228	228	391	391
R-squared	0.281	0.318	0.241	0.242

Notes: The estimation uses the linear probability model to predict whether the spouse is in the same workplace for migrant workers who married after migration. The sample includes individuals who married after migrating and either have no co-workers from the same town or have more than 30% of their co-workers from the same town. *Same-town co-worker* is an indicator for having over 30% co-workers from the same town. *Traditional* is an indicator of whether an individual hails from a province with traditional gender norms, as determined by responses to Q3. All specifications include the following controls as defined in Table 3: Education attainment, birth cohort fixed effects, occupation fixed effects, industry fixed effects, log salary, an indicator for whether it is the individual's first job, an indicator for whether the individual was referred to the job, the age at which the individual started the job, origin and destination province fixed effects. Robust standard errors are given in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

hometown.

In addition to the findings on the probability of early marriage presented in Table 5, examining a distinct outcome—whether the spouse is in the same workplace—reveals a similarly gender-asymmetric pattern, with the effect concentrated among females from regions with traditional gender norms. Both align with the possibility that female migrant workers are pressured to conform to traditional norms of early marriage from co-workers from the same hometown, leading them to marry someone from their workplace, regardless of whether the spouse is from the same hometown.

6.2 Selection

The concentration of same-origin co-workers may correlate with factors that affect the timing of marriage. For example, people who are less educated may be more likely to rely on the *tongxiang* network and, simultaneously, are more likely to marry early. Moreover, people who are more resourceful can join the *tongxiang* network and are also more capable of finding a marriage partner. It is also possible that even if male and female migrant workers sort into

²⁸Since same-sex marriage is not legalised in China, the matching hypothesis assumes that marriages involve a male and a female from the same origin. Therefore, the matching story would predict a similar positive association between same-town co-workers and the presence of a spouse in the same workplace for male migrants.

²⁹Such a scenario would require a similar positive correlation between the concentration of same-origin co-workers and the presence of a spouse in the same workplace for male migrants.

tongxiang networks similarly based on certain characteristics, these characteristics may influence their marriage behaviour differently, thus resulting in the gendered effect of social interactions.

To the extent that selection is based on observable characteristics, these can be explicitly accounted for. In Tables 3 and 5, I control for an extensive set of variables, including individual characteristics such as educational attainment and birth cohort, as well as job-related factors like occupation, industry, salary, whether it is the individual's first job, whether the job was obtained through referral, and the age at which the individual started the job, as well as origin and destination province fixed effects. The gender-differentiated effect of same-origin co-workers remains robust.

Unobservable individual characteristics may also influence both the presence of same-origin co-workers and the likelihood of early marriage. Specifically, one concern is that individuals with a private preference for early marriage may be more inclined to join workplaces with same-origin co-workers. This paper provides several strands of evidence to show that this scenario is unlikely. First, there is no association between early marriage and same-origin co-workers among women who married before migration (Table A3). If having same-origin co-workers reflected certain private preferences or tendencies for early marriage, we would expect to see a correlation between early marriage and having same-origin co-workers among women who were already married before meeting same-origin co-workers. However, this is not the case.³⁰ Second, while migrant workers' identification with their rural or farmer origin predicts early female marriage and serves as a plausible proxy for the preference for early marriage, it is not correlated with the concentration of co-workers from the same town (Table A10). Taken together, there does not appear to be sorting into a workplace with same-origin co-workers based on private preferences or tendencies for early marriage, conditional on observable factors.

7. Conclusion

This paper provides novel evidence that social interactions among rural-to-urban migrants can pressure individuals to conform to traditional gender norms. When women migrate from rural to urban areas, the economic conditions favouring early marriage change, yet their behaviour may still align with traditional norms due to social pressure. I find that the presence of individuals from the same hometown in the workplace significantly increases the probability of early marriage for female migrants, with no comparable effect observed for male migrants.

I show that the effect of same-origin co-workers on early marriage is concentrated among individuals from provinces with traditional gender norms. Further, examining a distinct outcome—whether the spouse is more likely to be in the same workplace—also reveals a gender-differentiated effect of same-origin co-workers, present exclusively for female migrants

³⁰Note that this contrasts with the findings related to same-origin friends in the current location or self-identification with a rural origin, both of which reflect a preference for early marriage. Table A5 and A9 show that among female migrants who married before migration, those who later form same-origin friendships or identified with their rural origin, tend to have married at younger age.

from provinces with traditional gender norms. This suggests that under social pressure from same-origin co-workers, female migrants are more likely to marry someone in the workplace, regardless of whether the spouse is from the same hometown.

These findings carry important policy implications: the economic independence of female migrant workers does not necessarily translate into autonomy over marriage timing, which may in turn affect their decisions regarding childbearing and career development. Policies aimed at reducing gender disparities through equal access to education and employment should take into account the persistent influence of traditional gender norms to enhance their effectiveness.

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8. Appendix

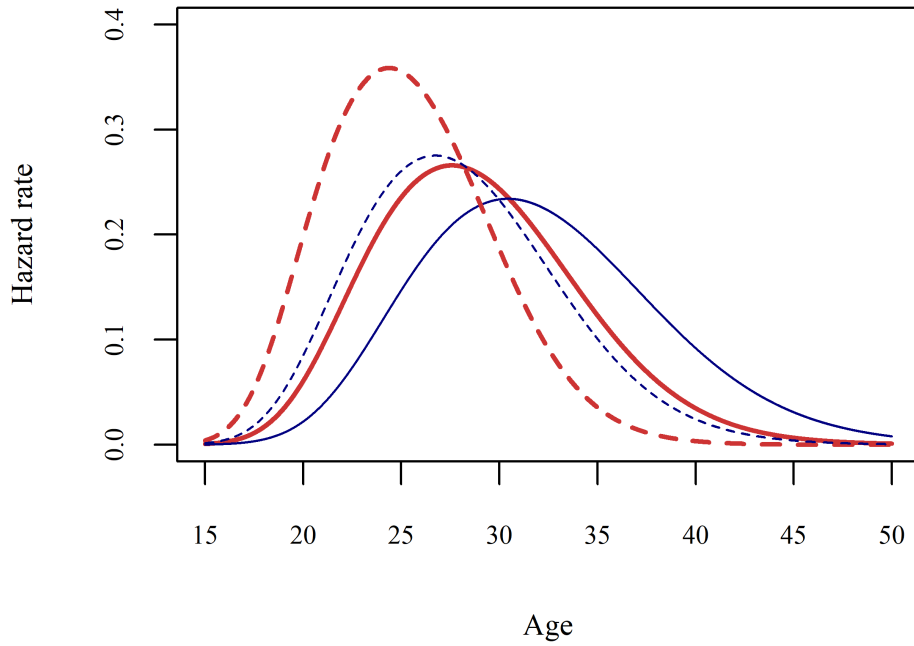


Figure A1: Marriage hazard rate in rural and urban China

— : urban female - - - : rural female — : urban male - - - : rural male

Notes: Based on the 2010 Chinese General Social Survey. *Rural* is defined as individuals who are born in rural regions and have never left. *Urban* is defined as individuals who are born with urban *hukou* and currently hold urban *hukou*.

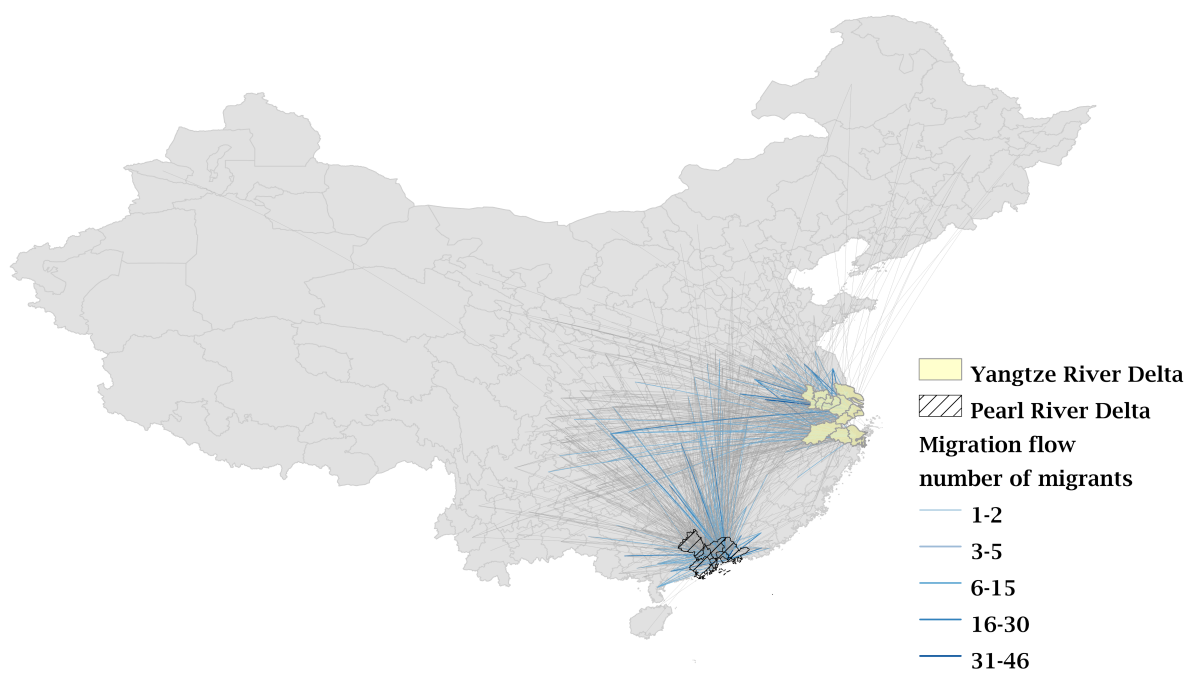


Figure A2: Migration flows to the Yangtze and Pearl River Deltas

Notes: The number of rural-to-urban migrant workers (prefecture to prefecture migration) calculated from the 2010 survey data on migrant workers. The data are representative of the regional distribution of migrants.

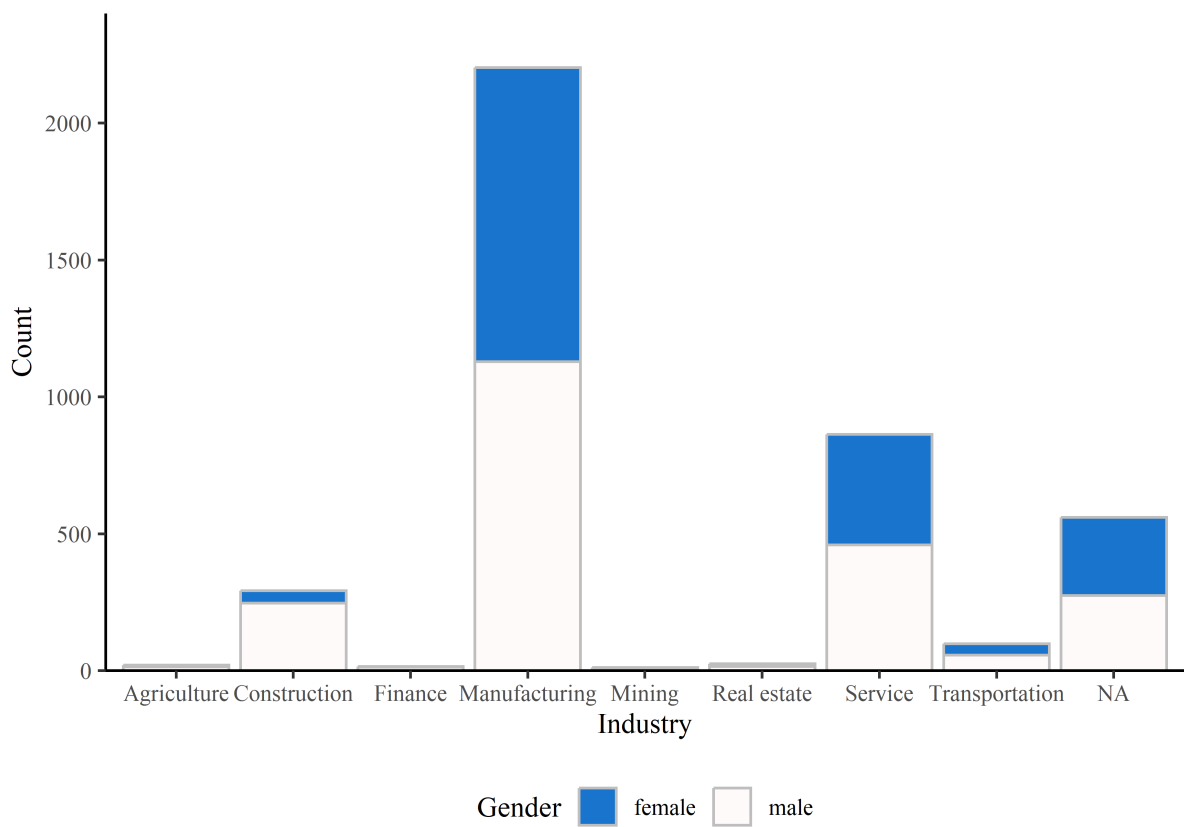
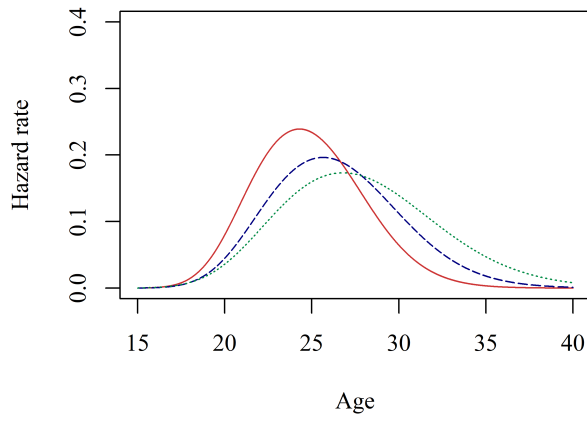
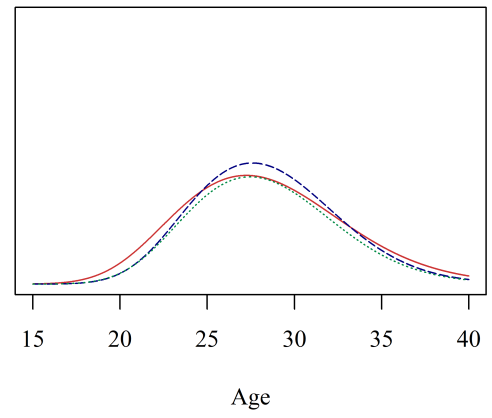


Figure A3: Industrial employment by gender

Source: the 2010 survey data on migrant workers.



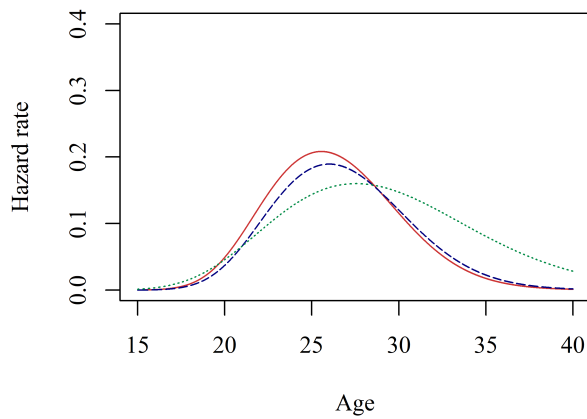
(a) Female



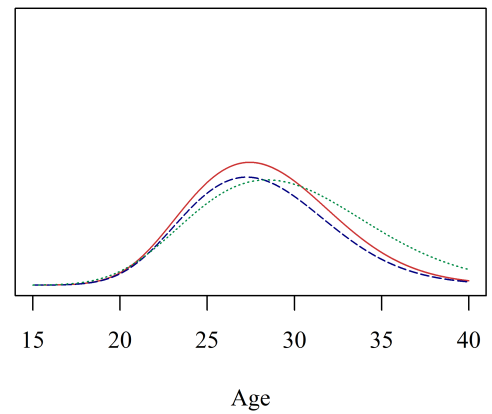
(b) Male

Figure A4: Estimated hazard rate: pc_county

- none of the co-workers come from the same county
- 0–30% of the co-workers come from the same county
- >30% of the co-workers come from the same county



(a) Female



(b) Male

Figure A5: Estimated hazard rate: pc_province

- none of the co-workers come from the same province
- 0–30% of the co-workers come from the same province
- >30% of the co-workers come from the same province

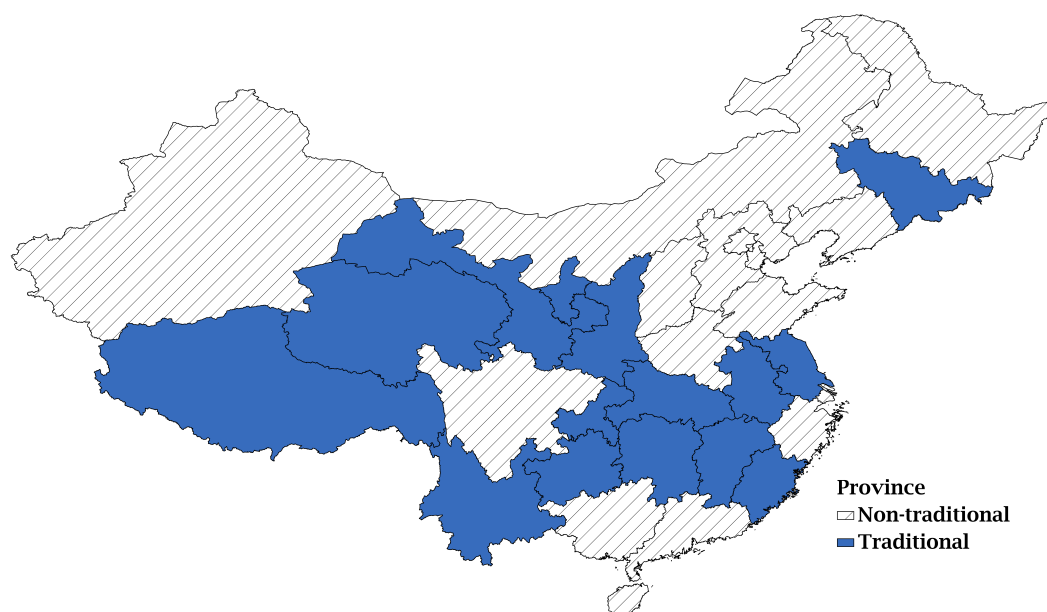


Figure A6: Traditional and non-traditional provinces

Notes: “Traditionalness” is defined by views of gender roles in society, as measured using responses from the 2010 Chinese General Social Survey. I calculate the median proportion of rural individuals in each province who agree with that “Men should prioritise their careers and women should prioritise their families.” Provinces with values above the median are categorised as traditional, while those below the median are classified as non-traditional.

Table A1: Perception of gender roles in rural and urban China

	Dep. var.: 1 (Traditional gender roles)
	(1)
Rural	0.292*** (0.016)
Male	0.058*** (0.019)
Rural × male	−0.074*** (0.024)
Constant	0.519*** (0.013)
Observations	6,408
R-squared	0.077

Notes: Based on the 2010 Chinese General Social Survey. The response variable is an indicator for whether the respondent agrees that men should prioritise career and women should prioritise family. The estimation uses the linear probability model to predict whether a respondent agrees with traditional gender roles. *Rural* is an indicator variable that equals one if the respondent was born and has stayed in the rural region and zero if the respondent was born and has stayed in the urban region. *Male* is an indicator that equals one if the respondent is male and zero otherwise. Robust standard errors are given in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A2: Proportion of same-town co-workers

	None	Very few <10%	A few 10%–20%	Some 20%–30%	Many 30%–50%	A lot >50%
Female, Married before migration (515)	0.46	0.31	0.08	0.07	0.05	0.04
Female, Married after migration (1258)	0.49	0.28	0.09	0.07	0.04	0.04
Male, Married before migration (538)	0.45	0.25	0.09	0.09	0.05	0.07
Male, Married after migration (1581)	0.49	0.25	0.10	0.07	0.04	0.06

Notes: Based on the 2010 survey on migrant workers. The table presents the concentration levels of same-town co-workers by gender and the timing of migration and marriage. The number of observations for each category in parentheses. A total of 179 individuals responded "do not know" to Q1, and 19 others have missing values.

Table A3: Marriage age and same-town co-workers—placebo analysis

Age	Probability of being married at or before a certain age					
	Female			Male		
	Baseline 0 same origin (1)	Difference >30% (2)	Difference >50% (3)	Baseline 0 same origin (4)	Difference >30% (5)	Difference >50% (6)
20	0.285	−0.039 (0.054)	−0.063 (0.228)	0.159	0.084* (0.044)	0.100* (0.056)
21	0.437	−0.036 (0.064)	−0.032 (0.190)	0.267	0.096* (0.055)	0.102 (0.070)
22	0.594	−0.028 (0.070)	0.011 (0.144)	0.398	0.093 (0.063)	0.087 (0.081)
23	0.731	−0.021 (0.070)	0.040 (0.104)	0.536	0.079 (0.066)	0.061 (0.086)
24	0.835	−0.016 (0.063)	0.046 (0.073)	0.662	0.059 (0.063)	0.034 (0.083)
25	0.904	−0.013 (0.051)	0.038 (0.047)	0.765	0.040 (0.055)	0.011 (0.076)
26	0.946	−0.011 (0.038)	0.026 (0.029)	0.841	0.025 (0.046)	−0.003 (0.065)
27	0.971	−0.010 (0.028)	0.016 (0.018)	0.895	0.014 (0.037)	−0.011 (0.054)
28	0.984	−0.008 (0.020)	0.009 (0.011)	0.930	0.008 (0.029)	−0.014 (0.044)
29	0.991	−0.007 (0.015)	0.005 (0.007)	0.953	0.004 (0.022)	−0.014 (0.035)
30	0.995	−0.006 (0.011)	0.003 (0.005)	0.967	0.002 (0.017)	−0.012 (0.028)

Notes: The estimation is based on the sample of individuals married before migration. The baseline group is migrants without any co-workers from the same town. The difference is the increase in probability relative to the baseline group for migrants with over 30% or 50% of the co-workers from the same town. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A4: Marriage age and friendship—married after migration

Age	Probability of being married at or before a certain age			
	Female		Male	
	Baseline (1)	Difference <i>Tongxiang</i> friend (2)	Baseline (3)	Difference <i>Tongxiang</i> friend (4)
20	0.058	0.049*** (0.015)	0.023	0.014** (0.007)
21	0.114	0.080*** (0.021)	0.053	0.022** (0.011)
22	0.194	0.111*** (0.028)	0.105	0.029* (0.015)
23	0.293	0.132*** (0.034)	0.180	0.033* (0.019)
24	0.399	0.140*** (0.038)	0.274	0.032 (0.024)
25	0.500	0.136*** (0.040)	0.377	0.028 (0.027)
26	0.587	0.125*** (0.040)	0.477	0.021 (0.029)
27	0.658	0.110*** (0.040)	0.566	0.015 (0.030)
28	0.712	0.096** (0.040)	0.640	0.010 (0.030)
29	0.752	0.084** (0.041)	0.698	0.007 (0.030)
30	0.781	0.074* (0.042)	0.741	0.006 (0.029)

Notes: The estimation is based on the sample of individuals married after migration. The baseline group is individuals whose best friend is not from the same hometown. The difference is the increase in probability relative to the baseline group for migrants whose best friend is from the same hometown. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A5: Marriage age and friendship—married before migration

Age	Probability of being married at or before a certain age			
	Female		Male	
	Baseline (1)	Difference <i>Tongxiang</i> friend (2)	Baseline (3)	Difference <i>Tongxiang</i> friend (4)
20	0.288	0.059 (0.037)	0.158	0.041 (0.029)
21	0.438	0.082** (0.039)	0.267	0.059* (0.035)
22	0.594	0.088** (0.040)	0.400	0.073* (0.039)
23	0.731	0.075** (0.038)	0.538	0.078* (0.040)
24	0.834	0.053* (0.032)	0.664	0.074** (0.037)
25	0.904	0.032 (0.025)	0.767	0.064** (0.032)
26	0.947	0.017 (0.018)	0.844	0.052** (0.026)
27	0.971	0.008 (0.012)	0.897	0.040** (0.020)
28	0.984	0.003 (0.008)	0.931	0.030* (0.016)
29	0.991	0.000 (0.006)	0.954	0.022* (0.012)
30	0.995	−0.001 (0.004)	0.968	0.017* (0.010)

Notes: The estimation is based on the sample of individuals married before migration. The baseline group is individuals whose best friend is not from the same hometown. The difference is the increase in probability relative to the baseline group for migrants whose best friend is from the same hometown. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A6: Marriage age and same-town co-workers—additional results

Age	Probability of being married at or before a certain age							
	Female				Male			
	Baseline 0 same origin (1)	Dif. 0-10% (2)	Dif. 10-30% (3)	Dif. >30% (4)	Baseline 0 same origin (5)	Dif. 0-10% (6)	Dif. 10-30% (7)	Dif. >30% (8)
20	0.070	0.004 (0.014)	0.008 (0.020)	0.069* (0.039)	0.027	0.000 (0.008)	−0.001 (0.009)	0.035* (0.019)
21	0.128	0.016 (0.020)	0.030 (0.027)	0.122** (0.048)	0.060	−0.001 (0.012)	0.000 (0.015)	0.054** (0.026)
22	0.207	0.033 (0.027)	0.063* (0.035)	0.174*** (0.057)	0.114	−0.003 (0.018)	0.003 (0.021)	0.072** (0.031)
23	0.302	0.047 (0.033)	0.101** (0.044)	0.208*** (0.065)	0.190	−0.007 (0.023)	0.007 (0.027)	0.083** (0.036)
24	0.404	0.052 (0.039)	0.131** (0.051)	0.217*** (0.071)	0.283	−0.013 (0.028)	0.014 (0.033)	0.086** (0.041)
25	0.503	0.048 (0.041)	0.148*** (0.055)	0.204*** (0.077)	0.385	−0.019 (0.032)	0.021 (0.037)	0.081* (0.046)
26	0.591	0.035 (0.043)	0.151*** (0.055)	0.177** (0.081)	0.483	−0.026 (0.035)	0.028 (0.040)	0.071 (0.049)
27	0.664	0.018 (0.043)	0.144*** (0.055)	0.145* (0.083)	0.571	−0.032 (0.037)	0.033 (0.040)	0.060 (0.050)
28	0.722	0.000 (0.044)	0.131** (0.054)	0.114 (0.084)	0.644	−0.036 (0.037)	0.035 (0.039)	0.050 (0.050)
29	0.767	−0.017 (0.046)	0.117** (0.053)	0.087 (0.084)	0.701	−0.039 (0.037)	0.036 (0.038)	0.042 (0.049)
30	0.801	−0.032 (0.048)	0.103* (0.052)	0.064 (0.084)	0.745	−0.041 (0.037)	0.036 (0.037)	0.037 (0.047)

Notes: The estimation is based on the sample of individuals married after migration. The baseline group consists of migrants without any co-workers from the same town. The reported differences represent the increase in probability relative to the baseline group for migrants with 0-10%, 10-30%, or more than 30% of co-workers from the same town. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A7: Marriage age and same-county co-workers—married after migration

Age	Probability of being married at or before a certain age					
	Female			Male		
	Baseline 0 same origin (1)	Difference >30% (2)	Difference >50% (3)	Baseline 0 same origin (4)	Difference >30% (5)	Difference >50% (6)
20	0.067	0.067* (0.035)	0.106 (0.117)	0.027	0.033* (0.017)	0.029 (0.020)
21	0.122	0.126*** (0.046)	0.180* (0.109)	0.058	0.050** (0.023)	0.044* (0.026)
22	0.199	0.187*** (0.056)	0.252** (0.101)	0.110	0.067** (0.028)	0.057* (0.032)
23	0.292	0.230*** (0.064)	0.298*** (0.095)	0.184	0.078** (0.033)	0.065* (0.038)
24	0.392	0.244*** (0.068)	0.310*** (0.091)	0.273	0.081** (0.037)	0.067 (0.046)
25	0.489	0.232*** (0.070)	0.295*** (0.088)	0.371	0.078* (0.041)	0.064 (0.053)
26	0.576	0.204*** (0.071)	0.263*** (0.085)	0.466	0.069 (0.044)	0.058 (0.058)
27	0.650	0.169** (0.072)	0.225*** (0.082)	0.552	0.059 (0.045)	0.051 (0.060)
28	0.708	0.135* (0.073)	0.189** (0.080)	0.624	0.050 (0.045)	0.046 (0.059)
29	0.754	0.105 (0.074)	0.158** (0.079)	0.681	0.043 (0.044)	0.043 (0.057)
30	0.788	0.080 (0.075)	0.132* (0.079)	0.724	0.039 (0.043)	0.041 (0.055)

Notes: The estimation is based on the sample of individuals married after migration. The baseline group is migrants without any co-workers from the same county. The difference is the increase in probability relative to the baseline group for migrants with over 30% or 50% of the co-workers from the same county. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A8: Marriage age and same-county co-workers—married before migration

Age	Probability of being married at or before a certain age					
	Female			Male		
	Baseline 0 same origin (1)	Difference >30% (2)	Difference >50% (3)	Baseline 0 same origin (4)	Difference >30% (5)	Difference >50% (6)
20	0.314	−0.064 (0.066)	−0.111 (0.113)	0.159	0.031 (0.038)	0.071 (0.051)
21	0.450	−0.025 (0.069)	−0.085 (0.108)	0.268	0.055 (0.047)	0.087 (0.063)
22	0.587	0.023 (0.066)	−0.039 (0.097)	0.400	0.077 (0.052)	0.092 (0.073)
23	0.708	0.055 (0.056)	0.003 (0.084)	0.537	0.091* (0.053)	0.086 (0.077)
24	0.805	0.064 (0.044)	0.026 (0.067)	0.662	0.092* (0.050)	0.072 (0.074)
25	0.874	0.057* (0.032)	0.031 (0.049)	0.764	0.082* (0.043)	0.056 (0.065)
26	0.921	0.043* (0.023)	0.027 (0.034)	0.840	0.068* (0.035)	0.041 (0.054)
27	0.951	0.030* (0.016)	0.020 (0.023)	0.892	0.053** (0.027)	0.029 (0.043)
28	0.970	0.020* (0.011)	0.013 (0.015)	0.927	0.040* (0.021)	0.020 (0.034)
29	0.981	0.013 (0.008)	0.008 (0.010)	0.950	0.030* (0.016)	0.015 (0.026)
30	0.988	0.008 (0.006)	0.005 (0.007)	0.964	0.023* (0.012)	0.011 (0.020)

Notes: The estimation is based on the sample of individuals married before migration. The baseline group is migrants without any co-workers from the same county. The difference is the increase in probability relative to the baseline group for migrants with over 30% or 50% of the co-workers from the same county. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A9: Marriage age and identity—married before migration

Age	Probability of being married at or before a certain age			
	Female		Male	
	Baseline Worker (1)	Difference Farmer (2)	Baseline Worker (3)	Difference Farmer (4)
20	0.303	−0.009 (0.059)	0.177	−0.012 (0.038)
21	0.423	0.049 (0.061)	0.293	−0.021 (0.047)
22	0.546	0.102* (0.060)	0.430	−0.030 (0.053)
23	0.660	0.129** (0.056)	0.571	−0.037 (0.058)
24	0.756	0.126** (0.050)	0.695	−0.041 (0.058)
25	0.831	0.106** (0.043)	0.794	−0.041 (0.054)
26	0.886	0.081** (0.037)	0.865	−0.038 (0.046)
27	0.925	0.058** (0.030)	0.914	−0.033 (0.038)
28	0.950	0.040* (0.024)	0.945	−0.028 (0.030)
29	0.967	0.027 (0.019)	0.964	−0.024 (0.024)
30	0.978	0.018 (0.015)	0.976	−0.020 (0.019)

Notes: The estimation is based on the sample of individuals married before migration. The baseline group is individuals who identify themselves as workers. The difference is the increase in probability relative to the baseline group for migrants who identify themselves as farmers. Bootstrap standard errors with 500 replicates are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table A10: Same-town co-workers and identity

	Dep. var.: Farmer identity			
	Female		Male	
	(1)	(2)	(3)	(4)
<i>pc_town</i>	0.013 (0.010)	0.002 (0.010)	0.011 (0.008)	−0.001 (0.009)
Education		−0.073*** (0.018)		−0.058*** (0.016)
Age at start of work		0.008** (0.004)		0.009*** (0.003)
First job		0.017 (0.029)		−0.030 (0.026)
Referral		0.020 (0.027)		0.006 (0.024)
Log(salary)		−0.102*** (0.038)		−0.072*** (0.032)
Occupation FE		✓		✓
Industry FE		✓		✓
Birth cohort FE		✓		✓
Origin and destination FE		✓		✓
Observations	1,426	1,182	1,776	1,524
R-squared	0.001	0.167	0.001	0.144

Notes: The response variable is an indicator for self-identification with farmer or rural origin. The main explanatory variable *pc_town* is a categorical variable that increases with the level of concentration of co-workers from the same town (0: none; 1: 0%–10%; 2: 10%–20%; 3: 20%–30%; 4: 30%–50%; 5: more than 50%). *Education* is a categorical variable that increases with the level of education (0: less than or equal to primary school; 1: junior high school; 2: senior high school; 3: more than high school). *Age at start of work* is the age at which the individual began working in their current job. *First job* is an indicator for whether the current job is the individual's first job. *Referral* indicates whether the job was obtained through a referral. *Log(salary)* calculates the natural logarithm of the individual's monthly salary. Robust standard errors are given in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.